MILITARY SPECIFICATION

# FIBERSCOPE, FIBER OPTIC (METRIC), GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements of fiber optic fiberscopes. Fiberscopes are used for the nondestructive inspection of equipment.

1.2 Classification. Fiberscopes covered by this specification are classified as specified in 1.2.1 through 1.2.4 (see 3.1).

1.2.1 Type. The type designation defines the type of fiberscope assembly.

R - Rigid fiberscope
F - Flexible fiberscope
T - Optional adapter tips

1.2.2 Style. The style designation defines the diameter of the fiberscope probe (see appendix, tables VII and VIII).

1.2.3 Class. The class designation defines the fiberscope design characteristics (see appendix, tables VII and VIII).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.4 Options. The options designation define those parameters that can vary in the fiberscope and that can be selected to tailor the fiberscope for its intended use (see appendix, tables IX, X, and XI).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| L-P-378 | - Plastic Sheet and Strip, Thin Gauge, Polyolefin. |
| QQ-S-781 | - Strapping, Steel, and Seals. |
| TT-I-735 | - Isopropyl Alcohol. |
| NNN-P-40 | - Paper, Lens. |
| PPP-B-585 | - Boxes, Wood, Wirebound. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |
| PPP-C-1752 | - Cushioning Material, Packaging, (Unicellular Polyethylene Foam, Flexible). |
| PPP-F-320 | - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes. |

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|---|---|
| MIL-P-116 | - Preservation, Methods of. |
| MIL-C-4150 | - Cases, Transit and Storage, Waterproof and Water-Vaporproof. |
| MIL-H-5606 | - Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance. |
| MIL-T-5624 | - Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST. |
| MIL-F-16884 | - Fuel, Naval Distillate. |
| MIL-L-17331 | - Lubricating Oil, Steam Turbine and Gear, Moderate Service. |
| MIL-H-17672 | - Hydraulic Fluid, Petroleum, Inhibited. |
| MIL-L-23699 | - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base. |
| MIL-F-24734/1 | - Fiberscope, Fiber Optic Type F, Style 1, Class 1, Options F. (Metric) |
| MIL-F-24734/2 | - Fiberscope, Fiber Optic Type R, Style 4, Class 1, Options S. (Metric) |
| MIL-P-26514 | - Polyurethane Foam, Rigid or Flexible, for Packaging. |



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STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-313 | - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities. |
| FED-STD-595 | - Colors. |

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|---|---|
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of U.S. Military Property. |
| MIL-STD-147 | - Palletized Unit Loads. |
| MIL-STD-454 | - Standard General Requirements for Electronic Equipment. |
| MIL-STD-470 | - Maintainability Program for Systems & Equipment. |
| MIL-STD-471 | - Maintainability Verification/Demonstration/ Evaluation. |
| MIL-STD-781 | - Reliability Testing for Engineering Development, Qualification, and Production. |
| MIL-STD-808 | - Finishes,, Materials and Processes for Corrosion Prevention and Control in Support Equipment. |
| MIL-STD-810 | - Environmental Test Methods and Engineering Guidelines. |
| MIL-STD-889 | - Dissimilar Metals. |
| MIL-STD-2073-1 | - DOD Materiel Procedures for Development and Application of Packaging Requirements. |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.3 Reliability (see 4.6.2). The fiberscope shall be designed to have a lower test mean-time-between-failures (MTBF $0_1$) of 500 hours and an upper mean-time-between-failures (MTBF $0_0$) of 1,110 hours.



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3.4 Materials. The units shall be constructed of materials that will not produce toxic, corrosive, or explosive by-products. Materials shall not have adverse effects upon operational or maintenance personnel under all operational and environmental conditions, nor cause degradation of equipment performance. Materials shall be non-magnetic.

3.4.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new. Products may be fabricated using raw materials produced from recovered bulk materials to the extent practicable if the intended use of the product is not jeopardized. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become part of a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of partially processed, assembled, used or rebuilt products are allowed under this specification.

3.4.2 Cadmium plating. Cadmium plating shall not be used.

3.4.3 Metals. Metals shall be corrosion-resistant. All exposed corrosion-resistant steel parts shall have a passivated surface that is compatible with external coatings or platings.

3.4.4 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless suitably finished to prevent electrolytic corrosion. Selection and protection of dissimilar metal combinations shall be in accordance with MIL-STD-889.

3.4.5 Nonmetallic materials. Nonmetallic materials used in the construction of fiberscopes shall not be degraded under the specified environmental conditions.

3.4.6 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.6).

3.5 Design and construction. The configuration and physical dimensions of the fiberscope shall be as specified herein and in the applicable specification sheet (see 3.1). The fiberscope shall be designed for simplicity of operation. Fiberscopes shall be designed and constructed to operate at an annual usage rate of 700 hours per year throughout their expected lifetime of 5 years. Physical parameters are defined in the appendix to this specification, figures 2 and 3, for rigid and flexible fiberscopes, respectively. Fiberscopes with the same part number shall be physically and functionally interchangeable.

3.5.1 Light source (see 4.6.1.1). The fiberscope shall be compatible with fiberscope light sources as specified (see 3.1).



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3.5.2 Flexible light guide cable (see 4.6.1.1). The light guide cable shall be not less than 1.8 meters long. It shall provide an illumination path from the light source to the fiberscope. The light-guide cable shall withstand bending into a 5 cm radius circle without incurring damage or loss of light transmission capabilities. The design shall provide an exterior sheath for fiber optic bundle protection. Unless otherwise specified (see 3.1), the light guide shall enter the fiberscope eyepiece/handle assembly from the bottom.

3.5.3 Light guide cable connector (plug) (see 4.6.1.1). The light guide cable connector shall provide a positive connection with the light guide cable socket. A force of 26.5 ± 2.2 newtons shall be required to mate, and a force of 43.6 ± 2.2 newtons shall be required to demate the light guide from the light source. The dimensions of the connector shall be in accordance with figure 1.

3.5.4 Eyecup (see 4.6.1.1). A soft, pliable, positive-mounted, rotatable, contoured eyecup void of all sharp surfaces shall be incorporated into the fiberscope to enhance visual acuity and adaptation to bright ambient light.

3.5.5 Eyepiece (see 4.6.1.1). The eyepiece shall be compatible with all ancillary equipment (see 3.5.19).

3.5.6 Articulation tip control (for type F fiberscopes only) (see 4.6.1.1). Articulation tip controls shall enable the viewer to maneuver the tip with one hand, leaving the other hand free to steady the viewing end and operate the focus mechanism. A locking device shall be provided which is capable of locking the tip at any position within the tip range of motion.

3.5.7 Stops (see 4.6.1.1). Fiberscopes and applicable accessories shall have mechanical stops on each manual control to prevent over-extension of movable elements such as the objective focus, eyepiece focus, and articulating tip adjustment.

3.5.8 Protective dust covers (see 4.6.1.1). Protective dust covers shall be provided for all external optical surfaces.

3.5.9 Protective sheathing (see 4.6.1.1). Exterior protection for the flexible probe of the type F fiberscope, the flexible light guide cable, and the rigid probe of the type R fiberscope shall be of corrosion-resistant stainless steel material. The material shall be oil- and water-proof, shall not fray or flake after extended use, and shall resist twisting along the probe axis.

3.5.10 Working length marking (see 4.6.1.1). Unless otherwise specified (see 3.1), the working length (see 6.5.8) of the fiberscope shall be permanently marked at 25.4-cm intervals.

3.5.11 Tip length (see 4.6.1.1). The tip length (see 6.5.7) shall be as specified (see 3.1).

3.5.12 Overall length (see 4.6.1.1). The overall length (see 6.5.5) shall be as specified (see 3.1).



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3.5.13 Working diameter (see 4.6.1.1). The working diameter shall be as specified (see 3.1).

3.5.14 Axial viewing angle (see 4.6.1.1). The axial viewing angle (see 6.5.3) shall be as specified (see 3.1).

3.5.15 Diopter adjustment (see 4.6.1.1). Ocular accommodation shall be provided by an adjustable diopter of minus 4 to plus 4 diopters.

3.5.16 Weight (see 4.6.3.1). The weight of the fiberscope shall be as specified (see 3.1).

3.5.17 Removable parts (see 4.6.1.1). The only removable parts entering the area being inspected shall be the fiberscope interchangeable tips (flexible fiberscopes only). The interchangeable tips shall incorporate a fail-safe tip locking device to prevent the loss of the tip in the inspection area.

3.5.18 Carrying case (see 4.6.1.1). The carrying case shall be in accordance with MIL-C-4150.

3.5.19 Ancillary equipment (see 4.6.1.1). The effective performance of fiberscopes often requires the use of accessory equipment. The accessories shall be as specified (see 3.1) and may consit of the following:

(a) Guide tubes to provide protection and serve as a guide for the fiberscope probe when inserted into the inspection area.
(b) Angle adapters to provide for easier access to the eyepiece where the viewing positions are awkward or tiresome, or where access restrictions may permit insertion of the fiberscope but prevent the observer from access to the eyepiece.

    (1) Off-set angle shall be as specified (see 3.1) and shall have a maximum tolerance of 2.5 degrees.
    (2) Angle adapter leg lengths shall be 10, 25, or 50 cm, or as specified (see 3.1).

(c) Straight extension adapters shall be 16, 25, or 50 cm, or as specified (see 3.1).
(d) Lecture scope that allows two observers to view the same object simultaneously, and is attachable to the eyepiece of type F fiberscopes. It shall be not less than 91-cm long with a diopter adjustment of minus 4 to plus 4.
(e) Camera adapters for attachment of a camera to the fiberscope. Attachment shall not change the image presentation as seen through the eyepiece during normal viewing. Off-the-shelf photographic components may be used and photographic equipment adapters shall be as specified (see 3.1).

    (1) Still camera adapters shall attach 35-mm, single-lens reflex cameras to the fiberscope eyepiece. The adapters shall allow for 360-degree rotation about the fiberscope eyepiece axis and have the proper interface to a specified "T" mount connector.


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(2) Video camera adapters shall attach a video camera to the fiberscope eyepiece. The adapter shall allow for 360-degree rotation about the fiberscope eyepiece axis with a standard video camera "C" mount connector.

(f) Attachment capability. All adapters shall attach to the fiberscope in less than two minutes without the use of tools.

3.5.20 Finish (see 4.6.1.1). The fiberscope and carrying case shall be coated to protect them from corrosion. The coatings used shall be in accordance with MIL-STD-808.

3.5.21 Color (see 4.6.1.1). The exterior of the carrying case shall be an orange-yellow color paint and have a non-gloss, smooth, satin finish in accordance with FED-STD-595.

3.5.22 Maintainability (see 4.6.3.2). The fiberscope shall have a mean-time-to-repair (MTTR) of 2.5 hours in accordance with MIL-STD-470, task 101, 203, and 301. Fiberscopes shall be designed to ensure parts replacement can be accomplished with the use of common hand tools and consumable cleaning materials. Maintenance and operating instructions shall be provided with each fiberscope.

3.6 Performance characteristics. The fiberscope shall display internal and external surfaces not normally accessible to the naked eye without substantial disassembly. It shall provide the capability of still photography and closed circuit television video (CCTV) monitoring. The fiberscope shall operate in any orientation and in any position in which it is placed.

3.6.1 Magnification (see 4.6.4.1). Fiberscope magnification (see 6.5.4) and angle of field shall be inversely related through the optics of the fiberscope. A specified angle of field shall result in a given magnification achievable by the fiberscope.

3.6.1.1 Eyepiece magnification (see 4.6.4.1). Magnification through the eyepiece shall be as specified (see 3.1).

3.6.2 Angle of field (see 4.6.4.2). The angle of field (see 6.6.1) shall be as specified (see 3.1).

3.6.3 Image curvilinear distortion (see 4.6.4.3). The curvilinear distortion shall be not greater than 2 percent of the actual image. Fiberscopes shall provide an image which meets this requirement when the distal tip (objective end) is within 25.4 ± 0.13 mm of the object being inspected.

3.6.4 Image voids (see 4.6.4.4). For type F fiberscopes, the allowable image void shall contain no more than 0.1 percent broken fibers. Single image voids shall be not greater than 0.0064 percent of the total area of the output image. The distance between single image voids shall be not less than 5 percent of the output image diameter. The total area of all the image voids shall be not greater than 0.5 percent of the total output image area.



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3.6.5 Lens adjustment (focus) (see 4.6.4.5). The fiberscopes shall have a lens adjustment that shall provide a clear image throughout the specified depth of focus.

3.6.6 Target resolution (see 4.6.4.6). The resolution, in line pairs per millimeter, (see 6.5.6), shall be as specified (see 3.1).

3.6.7 Light attenuation (see 4.6.4.7). When mated with a light source, the light loss through the fiberscope shall be not greater than 40 percent for type R and not greater than 65 percent for type F. No yellowing, discoloration, tinting, or selective filtering shall be produced by the fiberscope and the flexible light guide cable.

3.6.8 Working section radius (see 4.6.4.8). The minimum working section radius of type F fiberscopes shall be as specified (see 3.1). The fiberscope shall show no evidence of deterioration in image clarity or mechanical damage, or inoperability.

3.6.9 Flexibility (see 4.6.4.9). Type F fiberscopes shall be able to negotiate turns down to the specified minimum working section radius yet shall not deflect more than 5 cm when a 30 cm section is self-supporting.

3.6.10 Articulation angle (for type F fiberscopes only) (see 4.6.4.10). For the definition of articulation angle see 6.5.2 and figure 3 of the appendix. For 4-way articulation, the articulation angle shall be plus and minus 90 degrees in one plane, and plus and minus 110 degrees in the plane perpendicular. For 2-way articulation, the articulation angle shall be plus and minus 110 degrees in one plane. The articulation type shall be as specified (see 3.1).

3.7 Environmental requirements. The fiberscope shall meet all requirements specified under the following environmental conditions.

3.7.1 Environmental temperature extremes (see 4.6.5.1). The fiberscope shall be fully functional and show no evidence of physical degradation. The fiberscope environmental temperature range shall be as specified (see 3.1), in accordance with the ranges in table I.

TABLE I. Environmental temperature ranges.

| Range | Operating (°C) | Storage (°C) |
|---|---|---|
| 1 | -54 to +85 | -62 to +85 |
| 2 | -28 to +85 | -62 to +85 |

3.7.2 Thermal shock (see 4.6.5.2). The fiberscope shall be fully functional and show no evidence of physical degradation. The temperature extremes shall be the specified storage temperature extremes.



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3.7.3 Humidity (see 4.6.5.3). The fiberscope shall show no evidence of deterioration of parts or materials, loosening of finishes, physical distortion, corrosion of metals, entrapment of moisture, separation of bonded surfaces, or other physical damage.

3.7.4 Salt spray (corrosion) (see 4.6.5.4). The fiberscope shall show no evidence of salt penetration into the fiberscope housing nor evidence of corrosive effects that impairs its specified functionality.

3.7.5 Fluid immersion (see 4.6.5.5). The fiberscope shall reveal no evidence of swelling or softening of material, no loss of sealing capability, nor any other effects detrimental to the operation of the fiberscope.

3.7.6 Shock (see 4.6.5.6). The fiberscope shall show no evidence of physical damage that impairs its specified functionality.

3.7.7 Vibration (see 4.6.5.7). The fiberscope shall show no evidence of physical damage that impairs its specified functionality.

3.7.8 Fungus resistance. The fiberscope shall be in accordance with MIL-STD-454, requirement 4, for fungus-inert materials. Fiberscopes that are not in accordance with requirement 4 of MIL-STD-454 shall meet grade I classification requirements of MIL-STD-810, method 508 when tested as specified (see 4.6.5.8).

3.8 Human factors (see 4.6.1.1). Fiberscopes and accessories shall minimize operator fatigue with regard to assembling and operating the equipment over a 4-hour period under normal environmental conditions. Specifically, the fiberscopes shall:

(a) Permit simple and timely removal from its carrying case and assembly into an operating configuration without tools.
(b) Provide effective mechanical considerations to support, manipulate, and adjust as may be required for operation.
(c) Allow operation with one hand while wearing standard work gloves.
(d) Provide maximum optical performance by providing adjustable ocular accommodations and adaptation.

3.9 Safety characteristics (see 4.6.1.1). The fiberscope shall be designed so that operator error, component malfunction, and inclement weather, occurring simultaneously, will not cause electrical shock or other danger to the operator, other persons, or the equipment.

3.9.1 Safety. The handling and operation of the fiberscope shall not endanger personnel by direct contact or by long-term effects due to deterioration of any component within the fiberscope construction.

3.9.2 Detachability. All detachable components shall be easily attached and detached by means of connectors that incorporate a spring, detent, latch, or similar mechanical device to ensure positive attachment. The connectors shall prevent improper assembly and inadvertent disconnection.



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3.10 Transportability. The fiberscope operational performance specified herein shall not be affected following prolonged storage or shipment under environmental conditions for shipment and storage specified herein.

3.11 Identification marking (see 4.6.1.1). Parts, accessories, and assemblies shall be etched, stamped, or otherwise permanently and legibly marked with the part number, manufacturer's Commercial and Government Entity (CAGE), classification code, manufacturer's name, and date of manufacture in accordance with MIL-STD-130.

3.11.1 Nameplates. Metal nameplates with the part number of the fiberscope and each accessory shall be permanently attached to each carrying case.

3.12 Workmanship (see 4.6.1.1). Fiberscopes shall be free of defects, burrs, and sharp edges. Fiberscopes shall be cleaned of all foreign matter, such as metal chips, dirt particles, plastic powders and chips, and grease. Optical components shall be free of dust, dirt, grease, lint, or other foreign matter that could damage optical performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Inspection conditions shall be met as specified in the applicable test method for each inspection parameter, or as specified (see 3.1).

4.4 First article inspection. First article inspection shall consist of the tests listed in table II.



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4.4.1 Sample. A fiberscope sample shall be submitted for each fiberscope construction for which first article approval is desired. The sample submitted shall be seven fiberscopes of the same part number.

TABLE II. First article inspection.

| Inspection | Requirement | Test method | Sample size |
|---|---|---|---|
| Group I | | | |
| Visual and mechanical examinations | 3.5, 3.8, 3.9, 3.11, 3.12 | 4.6.1.1 | 7 |
| Weight | 3.5.16 | 4.6.3.1 | 1/ |
| Eyepiece magnification | 3.6.1.1 | 4.6.4.1 | 1/ |
| Angle of field | 3.6.2 | 4.6.4.2 | 1/ |
| Diopter adjustment | 3.5.15 | 4.6.1.1 | 1/ |
| Image curvilinear distortion | 3.6.3 | 4.6.4.3 | 1/ |
| Image voids | 3.6.4 | 4.6.4.4 | 1/ |
| Lens adjustment | 3.6.5 | 4.6.4.5 | 1/ |
| Target resolution | 3.6.6 | 4.6.4.6 | 1/ |
| Light attenuation | 3.6.7 | 4.6.4.7 | 1/ |
| Working section radius | 3.6.8 | 4.6.4.8 | 1/ |
| Flexibility | 3.6.9 | 4.6.4.9 | 1/ |
| Articulation angle | 3.6.10 | 4.6.4.10 | 1/ |
| Group II | | | |
| Environmental temperature extremes | 3.7.1 | 4.6.5.1 | 1 |
| Thermal shock | 3.7.2 | 4.6.5.2 | 2/ |
| Humidity | 3.7.3 | 4.6.5.3 | 2/ |
| Salt spray | 3.7.4 | 4.6.5.4 | 2/ |
| Fluid immersion | 3.7.5 | 4.6.5.5 | 1 |

See footnotes at end of table.



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TABLE II. First article inspection - Continued.

| Inspection | Requirement | Test method | Sample size |
|---|---|---|---|
| Shock | 3.7.6 | 4.6.5.6 | 3/ |
| Vibration | 3.7.7 | 4.6.5.7 | 3/ |
| Fungus resistance | 3.7.8 | 4.6.5.8 | 1 |
| Maintainability | 3.5.22 | 4.6.3.2 | 1 |
| Reliability | 3.3 | 4.6.2 | 3 |

1/ The same sample shall be used as in the visual and mechanical inspection.
2/ The same sample shall be used as in the environmental temperature extremes inspection.
3/ The same sample shall be used as in the fluid immersion inspection.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table II in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided into five lots for group II testing. The sample units shall then be subjected to the inspection for their particular lot. After completion of sample testing, all units shall be resubjected to group I testing. Any fiberscope failing any inspection shall not be subjected to further inspection.

4.4.3 Failures. One or more failures shall be sufficient cause for refusal to grant first article approval. When first article test units are taken from a larger lot, and the first article is disapproved, none of the units from that lot shall be delivered.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the inspections and tests specified for group A inspection (table III), group B inspection (table IV), and group C inspection (table V) (see 6.4).

4.5.1 Group A inspection. Group A inspections shall follow the order shown in table III.


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TABLE III. Group A inspection.

| Inspection | Requirement | Test method |
|---|---|---|
| Visual and mechanical | 3.5, 3.8, 3.9, 3.11, 3.12 | 4.6.1.1 |
| Weight | 3.5.16 | 4.6.3.1 |
| Eyepiece magnification | 3.6.1.1 | 4.6.4.1 |
| Angle of field | 3.6.2 | 4.6.4.2 |
| Diopter adjustment | 3.5.15 | 4.6.1.1 |
| Image curvilinear distortion | 3.6.3 | 4.6.4.3 |
| Image voids | 3.6.4 | 4.6.4.4 |
| Lens adjustment | 3.6.5 | 4.6.4.5 |
| Target resolution | 3.6.6 | 4.6.4.6 |
| Light attenuation | 3.6.7 | 4.6.4.7 |
| Articulation angle | 3.6.10 | 4.6.4.10 |

4.5.2 Group B inspection. Group B inspection shall consist of the inspections specified in table IV. Group B inspections shall be made on units that have passed the group A inspection.

TABLE IV. Group B inspection.

| Inspection | Requirement | Test method |
|---|---|---|
| Shock | 3.7.6 | 4.6.5.6 |
| Thermal shock | 3.7.2 | 4.6.5.2 |
| Working section radius | 3.6.8 | 4.6.4.8 |
| Flexibility | 3.6.9 | 4.6.4.9 |

4.5.3 Group C inspections. Group C inspections shall consist of the inspections in table V. Group C inspections shall have satisfactorily completed all group A and group B inspections. After completion of group C inspections, the samples shall be re-subjected to group A inspection.



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TABLE V. Group C inspection.

| Inspection | Requirement | Test method |
|---|---|---|
| Environmental temperature extremes | 3.7.1 | 4.6.5.1 |
| Humidity | 3.7.3 | 4.6.5.3 |
| Salt spray | 3.7.4 | 4.6.5.4 |
| Fluid immersion | 3.7.5 | 4.6.5.5 |
| Vibration | 3.7.7 | 4.6.5.7 |
| Fungus resistance | 3.7.8 | 4.6.5.8 |
| Maintainability | 3.5.22 | 4.6.3.2 |
| Reliability | 3.3 | 4.6.2 |

4.5.3.1 Disposition of inspection sample units. Sample units that have failed shall not be delivered.

4.6 Test methods.

4.6.1 General testing practice.

4.6.1.1 Visual and mechanical examinations. Unless otherwise specified (see 3.1), visual and mechanical examinations shall be performed using 5X magnification to verify that the design, construction, physical characteristics, dimensions, human factors, safety, identification marking, and workmanship are in accordance with the applicable requirements.

4.6.1.2 Standard test conditions. Unless otherwise specified (see 3.1), the following ambient conditions shall be used:

(a) Temperature: 25 plus 3, minus 5 degrees Celsius (°C).
(b) Relative humidity: 50 ± 30 percent.
(c) Absolute Pressure: 725 ± 75 mm of mercury.

4.6.1.3 Equivalent test methods. The use of equivalent test methods is allowed (see 6.2).

4.6.2 Reliability (see 3.3). Fiberscopes from the same inspection lot shall be used and tested in accordance with MIL-STD-781, appendix C, or as specified (see 3.1).



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4.6.3 Design and construction inspection.

4.6.3.1 Weight (see 3.5.16). The fiberscope weight shall be tested using scales of sufficient range and accuracy appropriate for the specified weight and tolerance.

4.6.3.2 Maintainability (see 3.5.22). Unless otherwise specified (see 3.1), maintainability verification test shall be conducted in accordance with MIL-STD-471, method 1.

4.6.4 Performance tests.

4.6.4.1 Magnification. An object target shall be moved to the maximum distance possible while still providing a focused image by the fiberscope of the entire target object. The magnification (M) shall be calculated as follows:

$$M = \frac{H1,\ \text{object size seen by eye}}{L1,\ \text{object-to-eye distance}}$$

$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \overline{H2,\ \text{object size seen through fiberscope}}$$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad L2,\ \text{object-to-fiberscope objective distance}$$

4.6.4.2 Angle of field. An object target shall be moved to the maximum distance possible while still providing a focused image by the fiberscopes of the entire target object. The view field ($\theta$) shall be calculated as follows:

$$\tan(\theta/2) = \frac{(H/2)}{L}$$

where: $\theta$ = angle of field in degrees
L = distance from object to objective lens
H = height of object target

4.6.4.3 Image curvilinear distortion. In making this measurement, the edge of a straight line shall be placed on the end points of the image, and photographed. The measured distortion shall not exceed the requirement of 3.6.3.

4.6.4.4 Image voids (see 3.6.4). Image voids shall be determined by photographing a white screen through the fiberscope. The photograph shall be magnified 25X.

4.6.4.5 Lens adjustment (focus) (see 3.6.5). An object target shall be moved to a maximum distance of 12.7 mm from the objective lens, and a focused image shall be obtained at this distance. Then the target shall be moved away from the objective lens in 25.4-mm increments to a minimum distance of 0.61 meters. A focused image shall be obtained at each increment.

4.6.4.6 Target resolution. Fiberscopes shall be rigidly fastened to a mounting fixture that will allow establishment of a fixed distance of 25.40 ± 0.13 mm between the objective lens and the USAF-1951 Resolution Target. A photograph shall be taken through the eyepiece and developed. The camera set-up shall result



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in no magnification of the image seen through the eyepiece. The values specified (see 3.6.6) shall be resolved when observed with the unaided eye through the eyepiece as well as on the photograph. The one-to-one photograph shall be magnified 25X.

4.6.4.7 Light attenuation (see 3.6.7). A light source compatible with the item under test shall be turned on and allowed to warm up. After warm-up, the total radiant power, in watts, and spectral density, in watts per hertz, of the light emitted from the light source shall be measured at the point where the light guide plug terminates in the light source socket. The fiberscope light guide shall be connected to the light source and the total radiant power and spectral density emitted at the fiberscope distal end shall be measured.

4.6.4.8 Working section radius. The type F flexible probe shall be bent to the minimum working section radius specified (see 3.6.8). The fiberscope shall be visually examined for deterioration in image clarity, mechanical damage, or inoperability.

4.6.4.9 Flexibility. The type F flexible probe shall be placed flat on a horizontal table with the last 30 cm extending beyond the edge of the table. The heaviest tip intended for use shall be installed on the distal end for this test. Deflection of the end of the bundle shall not exceed that specified (see 3.6.9).

4.6.4.10 Articulation angle (see 3.6.10). The type F fiberscope tip shall be bent, using the articulation controls, the specified number of degrees and the articulation angle shall be verified.

4.6.5 Environmental tests.

4.6.5.1 Environmental temperature extremes (see 3.7.1). The fiberscope shall be tested in accordance with MIL-STD-810, method 501, procedure II, at the specified high operating temperature extreme. The fiberscope shall be tested in accordance with MIL-STD-810, method 502, procedure II, at the specified low operating temperature extreme. The test duration shall be 96 hours. At the completion of the test, the fiberscope shall be visually examined in accordance with 4.6.1.1.

4.6.5.2 Thermal shock (see 3.7.2). Fiberscopes shall be tested in accordance with MIL-STD-810, method 503 for "Basic Hot (A2)", air deliverable, and "Cold (C2)" conditions. At the completion of the test, the fiberscope shall be visually examined in accordance with 4.6.1.1.

4.6.5.3 Humidity (see 3.7.3). Fiberscopes shall be tested in accordance with MIL-STD-810, method 507, procedure II, nonhazardous/induced. The relative humidity shall be 90 percent, and the temperature range shall be the specified operating temperature range. At the completion of the test, the fiberscope shall be visually examined in accordance with 4.6.1.1.

4.6.5.4 Salt spray (corrosion) (see 3.7.4). Fiberscopes shall be tested in accordance with MIL-STD-810, method 509, for resistance to a 5 ± 1 percent concentration salt fog environment for a 48-hour period. At the completion of the test, the fiberscope shall be visually examined in accordance with 4.6.1.1.



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4.6.5.5 Fluid immersion (see 3.7.5). Fiberscopes shall be tested for resistance to fluids specified in table VI. A portion of the fiberscope probe outer cover, adhesive areas, and the objective lens shall be covered with each fluid listed in table VI. Each type fluid shall remain on each area for a period of 48 hours. At the completion of the test, the fiberscope shall be visually examined in accordance with 4.6.1.1.

TABLE VI. Immersion test fluids.

| Fluids | Specification | Test temperature (°C) |
|---|---|---|
| Fuel oil | MIL-F-16884 | 33 - 37 |
| Turbine fuel (JP-4 and JP-5) | MIL-T-5624 | 20 - 25 |
| Isopropyl alcohol | TT-I-735 | 20 - 25 |
| Hydraulic fluids | MIL-H-5606<br>MIL-H-17672 | 48 - 50 |
| Lubricating oils | MIL-L-17331<br>MIL-L-23699 | 73 - 77 |
| Coolant 1/ | - | 20 - 25 |
| Seawater | - | 20 - 25 |
| Tap water | - | 20 - 25 |

1/ Monsanto Coolanol 25 or equivalent.

4.6.5.6 Shock (see 3.7.6). Fiberscopes and accessories, in their carrying cases, shall be tested in accordance with MIL-STD-810, method 516, procedure IV. The fiberscope and accessories, after exposure to the designated shock conditions, shall be removed from the carrying case and visually examined in accordance with 4.6.1.1.

4.6.5.7 Vibration (see 3.7.7). Fiberscopes and accessories, in their carrying cases, shall be tested in accordance with MIL-STD-810, method 514, procedure I, categories 1 and 3. At the completion of the test, the fiberscope and accessories shall be removed from the carrying case and visually examined in accordance with 4.6.1.1.

4.6.5.8 Fungus resistance (see 3.7.8). Fiberscopes and carrying cases that do not meet the requirements of fungus-inert materials in accordance with MIL-STD-454, requirement 4, shall be tested by exposure to fungus in accordance with MIL-STD-810, method 508.



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4.7 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance requirements of MIL-P-116. The sampling and inspection of the packaging and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5 and the marking requirements of MIL-STD-129.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.9.)

5.1 Preservation. Unless otherwise specified (see 6.2), the preservation shall be level A.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The fiberscope and accessories shall be cleaned and dried in accordance with MIL-P-116, process C-1.

5.1.1.2 Preservation application. Preservatives shall not be used.

5.1.1.3 Unit packs. Each fiberscope shall be individually packed in accordance with MIL-P-116, method IA8, specified herein ensuring compliance with applicable requirements of that specification. Storage or transportation containers shall conform to the requirements of PPP-B-636, weather-resistant, with cushioning in accordance with MIL-P-26514 as required to preclude shock and vibration damage. The exposed optical surfaces on each assembly shall be covered with two layers of neutral lens tissue, NNN-P-40, type F, secured in position with pressure-sensitive tape. The fiberscope shall be secured to prevent vibration or shock damage. Either of two pieces of PPP-F-320 or PPP-C-1752 secured with pressure-sensitive tape shall be placed around the fiberscope. The fiberscope shall then be wrapped in accordance with L-P-378 and cushioned within the container with MIL-P-26514.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The packaged fiberscopes shall be packaged in containers conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specifications except the strapping shall conform to QQ-S-781, type R or IV, finish A. When the gross weight exceeds 90.91 kg or the container length and width is 121.9 by 60.96 cm or more and the weight exceeds 45.5 kg, 7.62 by 10.16 cm skids (laid flat) shall be applied in accordance with the requirements of the container specifications.

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5.2.2 Level B. The packaged fiberscope shall be packaged in fiberboard containers conforming to PPP-B-636, class weather-resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joints with waterproof packaging tape or box closure and sealing tape 5 cm minimum width.

5.2.3 Level C. The level C packaging for the packaged fiberscope shall conform to the MIL-STD-2073-1 requirements for this level.

5.2.4 Palletized loads. When specified (see 6.2), palletized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. Unless otherwise specified (see 6.2), the pallet pattern shall be in accordance with MIL-STD-147.

5.3 Marking.

5.3.1 Levels A, B, and C. In addition to any special or other identification marking required by the contract or order (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cubage consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fiberscope shall be designed to provide viewing of the internal areas of a wide variety of installed and uninstalled equipment during the performance of nondestructive inspection.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of the specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
(c) Whether first article inspection is required (see 3.2).
(d) Equivalent test methods, if other than as specified herein (see 4.6.1.3).



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(e) Level of packaging required (see 5.1 and 5.2).
(f) When loads are to be palletized (see 5.2.4).
(g) When pallet pattern is other than as specified (see 5.2.4).
(h) Marking requirements (see 5.3).
(i) Applicable specification sheet number, title, and date.
(j) Applicable part number (see 6.7).
(k) Quantity of fiberscopes required.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ____ production items, a standard production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Quality conformance. Quality conformance inspections require contractual definition of the overall test program, including sample sizes and lot sizes, if appropriate (see 4.5).

6.5 Definitions.

6.5.1 Angle of field. The greatest angle, in degrees, between two rays coming from the object through the objective lens into the optical system. Also known as field of view.

6.5.2 Articulation angle. Articulation angle is the angle between the axis of the articulating section of the fiberscope at its maximum deflection from the straight line working length position and the axis of the fiberscope tip at the undeflected position.

6.5.3 Axial viewing angle. Axial viewing angle is the angle by which the axis of view field deviates from the nominal straight forward axis of the working section.

6.5.4 Magnification. Magnification is the ratio of the apparent size of an object seen through the fiberscope to that of the size of the same object viewed directly by the unaided eye, with the object-to-fiberscope distance equal to the object-eye-distance.

6.5.5 Overall length. Overall length is equal to the working length, tip length, and eyepiece assembly combined.


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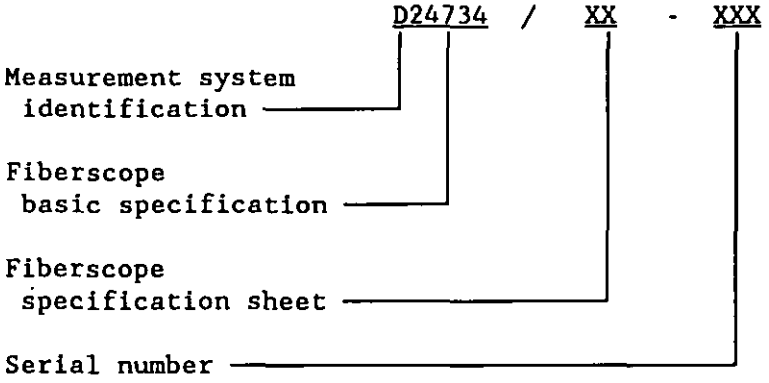
6.5.6 Resolution. Resolution is the resolving power of type R and type F fiberscopes specified in terms of line pairs per millimeter and is the measure of the ability of a lens or optical system to form separate and distinct images of two objects close together. Resolution can be measured by applying the USAF-1951 *Standard Resolution Chart. The resolution of a type F fiberscope is determined by* the number of fibers, as well as the number of broken fibers which cannot convey image information. The maximum acceptable number of broken fibers should be stated in terms of the percentage of total fibers.

6.5.7 Tip length. Tip length is the distance between the center of the objective window and the objective end (distal tip) of the flexible fiberscope. This distance determines how close to the bottom of the cavity one can observe an image when the fiberscope is completely inserted. The tip length should be compatible with the diameter of the probe.

6.5.8 Working length. The working length is the distance between the distal end of the hand-held portion and the center of the objective windows along the optical axis (see figures 2 and 3).

6.6 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Part or Identifying Number (PIN). Fiberscopes should contain only the following:




Examples:

D24734/01-001
D24734/01-002



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6.8 Part designator. A part designator, if specified (see 3.1), should include classifications (see 1.2) as follows:

(a) Type (see 1.2.1).
(b) Style (see 1.2.2).
(c) Class (see 1.2.3).
(d) Options (see 1.2.4).

Example: F11D.

6.9 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

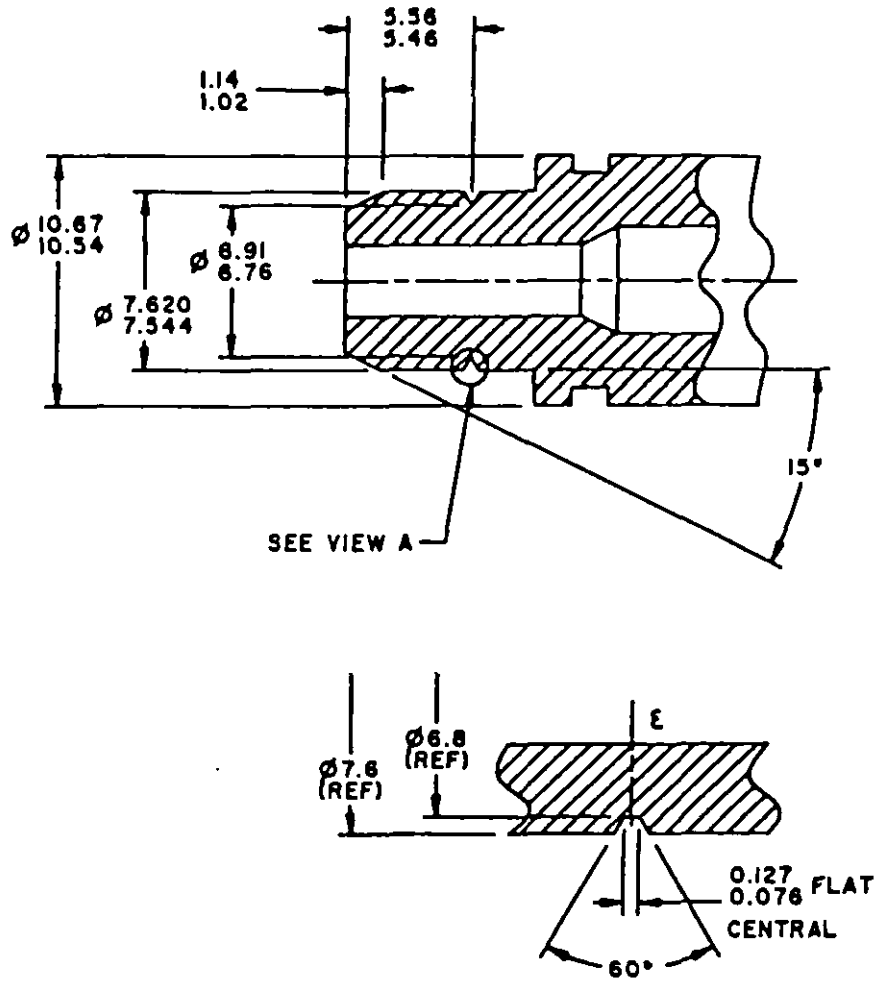
6.10 Subject term (key word) listing.

Borescope
Endoscope
Flexible fiberscope
Inspection scope

Review activities:
Navy - AS, EC, YD

Preparing activity:
Navy - SH
(Project 6035-N001)


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NOTES:

1. Dimensions are in millimeters.

2. All machined holes:
   Thru 3.18 mm +0.13 mm
   -0.05 mm

   Over 3.18 mm
   Thru 38.10 mm +0.30 mm
   -0.08 mm

   Over 38.10 mm +0.51 mm
   -0.10 mm

3. All surfaces 3175/$\mu$ mm.

4. Angles plus or minus 2 degrees.

FIGURE 1. Light guide cable connector (plug).



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# APPENDIX

## DESIGN CHARACTERISTICS FOR FIBERSCOPES

10. SCOPE

10.1 Scope. This appendix shows the design characteristics and physical parameters for the flexible fiberscope, rigid fiberscope, and the optional tips. This appendix forms a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DESIGN CHARACTERISTICS

30.1 Design characteristics. The design characteristics for flexible and rigid fiberscopes are shown in tables VII and VIII. The design characteristics for optional flexible fiberscope tips are shown in table IX.

30.2 Options. The design characteristics for options of the flexible and rigid fiberscopes are shown in tables X and XI.

30.3 Drawings. The fiberscope physical parameters are shown on figures 2 and 3.



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TABLE VII. Flexible fiberscope design characteristics.

| Style | Class | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 4.0 | 0.6 | 12.4 | 0.7 | 58.4 | 152.4 | 16.6 | 2 | 3 | 5.05 |
| 2 | 1 | 6.0 | 0.8 | 15.2 | 1.1 | 40.7 | 152.4 | 14.6 | 2 | 3 | 5.05 |
| | 2 | | 1.3 | 15.2 | 1.6 | 50.8 | 152.4 | 14.6 | 2 | 3 | 5.05 |
| | 3 | | 1.8 | 15.2 | 2.1 | 58.4 | 152.4 | 14.6 | 2 | 3 | 5.05 |
| 3 | 1 | 7.5 | 1.8 | 17.3 | 2 | 63.5 | 259.1 | 13.1 | 2 | 3 | 5.05 |
| 4 | 1 | 8.0 | 0.8 | 18 | 1.1 | 61 | 259 | 12.7 | 2 | 3 | 5.05 |
| | 2 | | 1.3 | 18 | 1.6 | 61 | 259 | 12.7 | 2 | 3 | 5.05 |
| | 3 | | 1.4 | 18 | 1.6 | 61 | 259 | 12.7 | 2 | 3 | 5.05 |
| | 4 | | 1.8 | 18 | 3.1 | 61 | 259 | 12.7 | 2 | 3 | 5.05 |
| | 5 | | 2.7 | 18 | 3.1 | 61 | 259 | 12.7 | 2 | 3 | 5.05 |
| 5 | 1 | 11.0 | 0.8 | 20.3 | 1.1 | 101.6 | 259.1 | 9.6 | 2 | 3 | 5.05 |
| | 2 | | 1.3 | 20.3 | 1.6 | 101.6 | 259.1 | 9.6 | 2 | 3 | 5.05 |
| | 3 | | 1.8 | 20.3 | 2.1 | 101.6 | 259.1 | 9.6 | 2 | 3 | 5.05 |
| | 4 | | 2.7 | 20.3 | 3.0 | 101.6 | 259.1 | 9.6 | 2 | 3 | 5.05 |
| | 5 | | 2.7 | 20.3 | 3.1 | 101.6 | 259.1 | 9.6 | 2 | 3 | 5.05 |
| 6 | 1 | 13.0 | 6.1 | 30.5 | 6.4 | 101.6 | 259.1 | 7.5 | 2 | 3 | 5.05 |

A = Working diameter (mm).
B = Working length (meters) (minimum length, plus 2.5 cm).
C = Tip length (mm).
D = Overall length (meters) (plus or minus 5.0 cm).
E = Minimum working section radius (mm).
F = Focus range (mm).
G = Objective distance (mm).
H = USAF group number.
I = USAF target number.
J = Line pairs per mm.



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TABLE VIII. Rigid fiberscopes design characteristics.

| Style | Class | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 3.5 | 27.9 | 38 | 17 | 4 | 1 | 16 |
| | 2 | | 30.5 | 41 | 17 | 4 | 1 | 16 |
| 2 | 1 | 4.0 | 28 | 38 | 16.6 | 2 | 6 | 7.12 |
| | 2 | | 35.6 | 46 | 16.6 | 2 | 6 | 7.12 |
| 3 | 1 | 4.5 | 40.64 | 51 | 16 | 4 | 1 | 16 |
| | 2 | | 27.94 | 38 | 16 | 4 | 1 | 16 |
| | 3 | | 30.48 | 41 | 16 | 4 | 1 | 16 |
| 4 | 1 | 5.0 | 34.8 | 45 | 15.8 | 4 | 1 | 16 |
| | 2 | | 15.8 | 17 | 15.8 | 4 | 1 | 16 |
| 5 | 1 | 5.5 | 35.6 | 46 | 15.08 | 4 | 1 | 16 |
| | 2 | | 45.72 | 47 | 15.08 | 4 | 1 | 16 |
| | 3 | | 33.02 | 34 | 15.08 | 4 | 1 | 16 |
| | 4 | | 48 | 58 | 15.08 | 4 | 1 | 16 |
| | 5 | | 76 | 77 | 15.08 | 4 | 1 | 16 |
| 6 | 1 | 6.5 | 55.8 | 66 | 14 | 4 | 1 | 16 |
| 7 | 1 | 8.0 | 27 | 38 | 12.5 | 4 | 1 | 16 |
| | 2 | | 5.7 | 7.62 | 12.5 | 4 | 1 | 16 |
| | 3 | | 25.4 | 36 | 12.5 | 4 | 1 | 16 |
| | 4 | | 33.02 | 44 | 12.5 | 4 | 1 | 16 |
| | 5 | | 88.9 | 99 | 12.5 | 4 | 1 | 16 |
| 8 | 1 | 10.0 | 38.1 | 48.2 | 10.5 | 4 | 1 | 16 |
| | 2 | | 33.1 | 43.1 | 10.5 | 4 | 1 | 16 |
| | 3 | | 45.7 | 55.8 | 10.5 | 4 | 1 | 16 |
| | 4 | | 88.9 | 98.9 | 10.5 | 4 | 1 | 16 |
| | 5 | | 37.1 | 47.2 | 10.5 | 4 | 1 | 16 |
| | 6 | | 48 | 58.1 | 10.5 | 4 | 1 | 16 |
| | 7 | | 59 | 69.1 | 10.5 | 4 | 1 | 16 |
| | 8 | | 27 | 37.1 | 10.5 | 4 | 1 | 16 |
| | 9 | | 69.1 | 79.2 | 10.5 | 4 | 1 | 16 |
| | 10 | | 29.5 | 39.5 | 10.5 | 4 | 1 | 16 |

A - Working diameter (mm).
B - Working length (cm) (minimum length, plus 2.5 cm).
C - Overall length (cm) (plus or minus 5.0 cm).
D - Objective distance (mm).
E - USAF Group number.
F - USAF Target number.
G - Line pairs per mm (minimum requirement).



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TABLE IX. Optional flexible fiberscope tips.

| Style | Option | A | B | C | D | E | F |
|---|---|---|---|---|---|---|---|
| 4 | 1 | 8.0 | 20 | 0 | 10 | 28 | 12.7 |
| | 2 | | 20 | 90 | 10 | 31 | 12.7 |
| | 3 | | 40 | 0 | 5 | 28 | 12.7 |
| | 4 | | 40 | 90 | 5 | 31 | 12.7 |
| | 5 | | 60 | 0 | 3 | 28 | 12.7 |
| | 6 | | 60 | 90 | 3 | 31 | 12.7 |
| | 7 | | 80 | 0 | 2 | 28 | 12.7 |
| | 8 | | 80 | 90 | 2 | 31 | 12.7 |
| 5 | 1 | 11.0 | 20 | 0 | 13 | 35.1 | 9.6 |
| | 2 | | 20 | 90 | 13 | 39.5 | 9.6 |
| | 3 | | 40 | 0 | 5 | 31.5 | 9.6 |
| | 4 | | 40 | 90 | 5 | 39.5 | 9.6 |
| | 5 | | 60 | 0 | 3 | 34.7 | 9.6 |
| | 6 | | 60 | 90 | 3 | 39.5 | 9.6 |
| | 7 | | 80 | 0 | 2 | 35.5 | 9.6 |
| | 8 | | 80 | 90 | 2 | 39.5 | 9.6 |

A - Working diameter (mm).
B - Angle of field (degrees).
C - Axial viewing angle (degrees).
D - Depth of focus (mm).
E - Tip length (mm).
F - Objective distance (mm).

TABLE X. Flexible fiberscope options.

| Options | X | Y |
|---|---|---|
| A | 0 | 20 |
| B | 90 | 20 |
| C | 0 | 30 |
| D | 90 | 30 |
| E | 0 | 40 |
| F | 90 | 40 |
| G | 0 | 50 |
| H | 90 | 50 |
| I | 0 | 60 |
| J | 90 | 60 |

X - Axial viewing angle (degrees).
Y - Angle of field (degrees) (plus or minus 5 degrees).


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TABLE XI. Rigid fiberscope options.

| Options | X | Y |
|---|---|---|
| A | 0 | 20 |
| B | 0 | 30 |
| C | 0 | 40 |
| D | 0 | 50 |
| E | 0 | 60 |
| F | 0 | 70 |
| G | 0 | 80 |
| H | 0 | 90 |
| I | 55 | 20 |
| J | 55 | 30 |
| K | 55 | 40 |
| L | 55 | 50 |
| M | 55 | 60 |
| N | 55 | 70 |
| O | 55 | 80 |
| P | 55 | 90 |
| Q | 90 | 20 |
| R | 90 | 30 |
| S | 90 | 40 |
| T | 90 | 50 |
| U | 90 | 60 |
| V | 90 | 70 |
| W | 90 | 80 |
| X | 90 | 90 |
| Y | 115 | 20 |
| Z | 115 | 30 |
| AA | 115 | 40 |
| BB | 115 | 50 |
| CC | 115 | 60 |
| DD | 115 | 70 |
| EE | 115 | 80 |
| FF | 115 | 90 |

X - Axial viewing angle (degrees) (plus or minus 0 degrees for 0 and 90 degree viewing angles; plus or minus 5 degrees for 55 and 115 degree viewing angles).
Y - Angle of field (degrees) (plus or minus 5 degrees).



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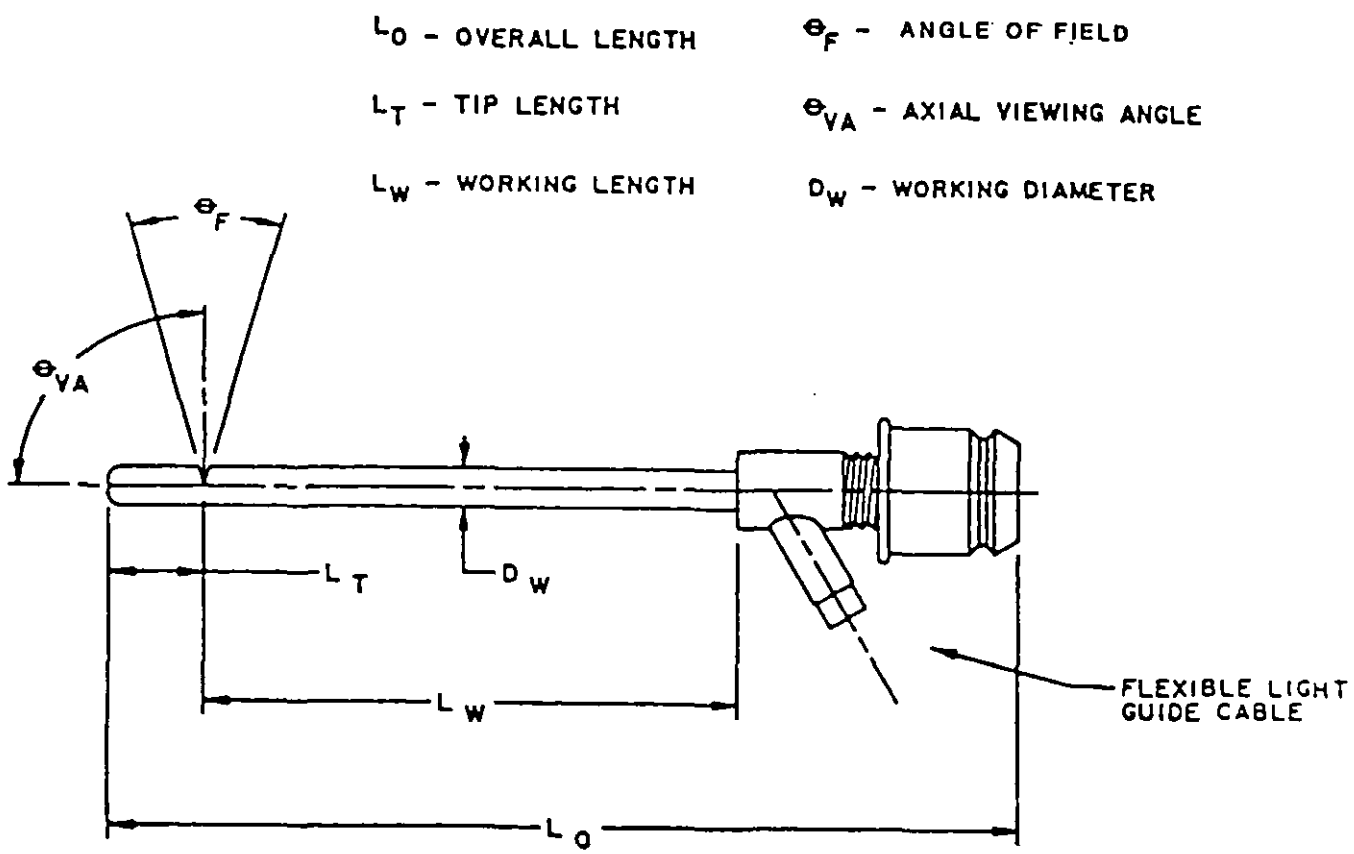



FIGURE 2. Rigid fiberscope probe physical parameters.



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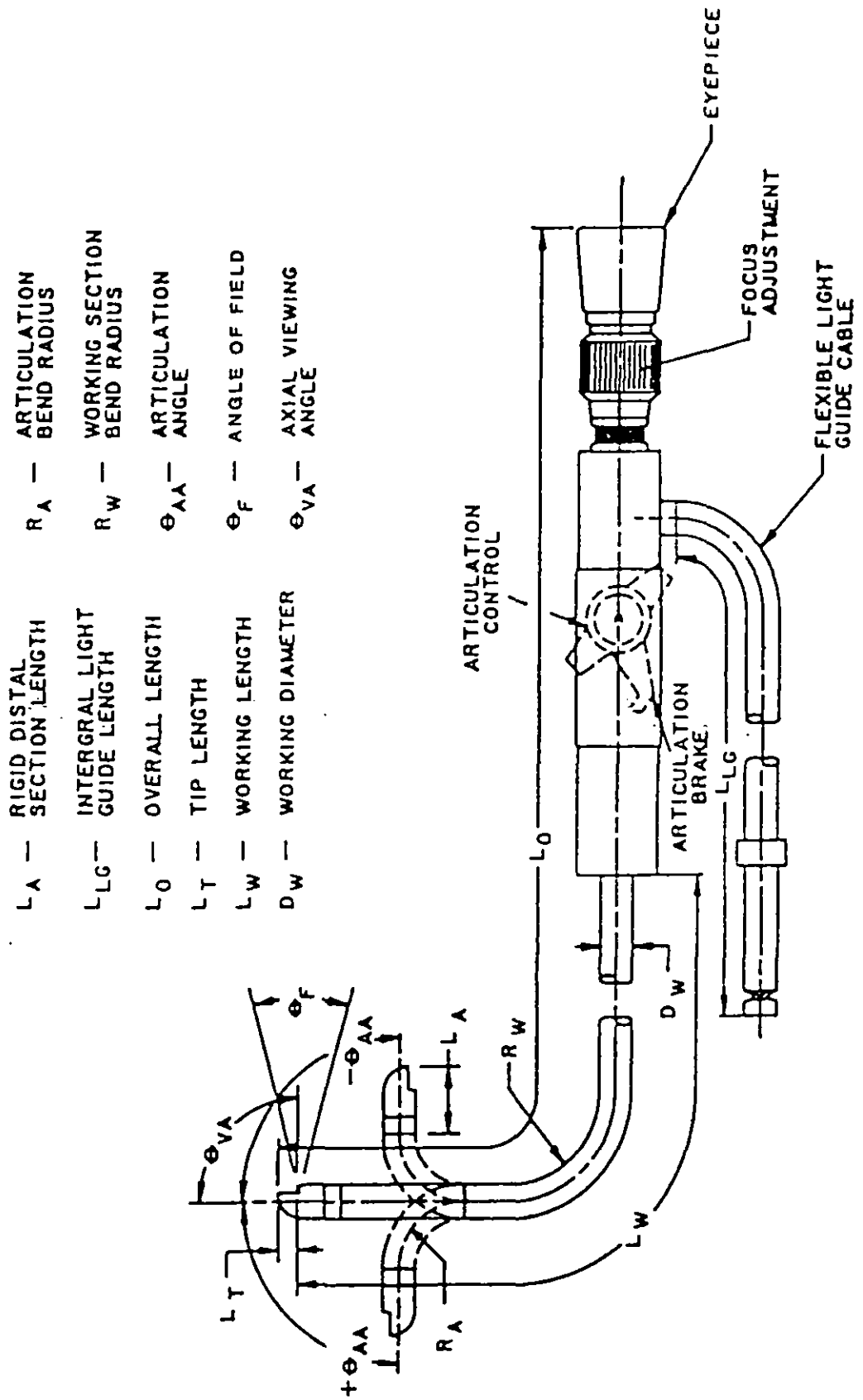



FIGURE 3. Flexible fiberscope probe physical parameters.



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|---|---|
| MIL-F-24734(NAVY) | FIBERSCOPE, FIBER OPTIC (METRIC) GENERAL SPECIFICATION FOR |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: |
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